Abir Zahraoui



AI Engineering | Machine Learning | Data Science | Generative AI

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PROFESSIONAL SUMMARY

Master's student in Artificial Intelligence and Connected Objects for Automotive (Université Ibn Tofail, 2025–2027) and holder of a Licence d'Excellence in Artificial Intelligence and Data Engineering. Technical experience in Machine Learning, data analysis, Generative AI, NLP, and AI automation, with projects in natural-language data analytics, predictive modeling, cybersecurity, and ROS2-based intelligent systems. Seeking a final-year internship (PFE) in AI Engineering, Machine Learning, Data Science, or Applied AI.

EDUCATION

Master d'Excellence — Intelligence Artificielle et Objets Connectés pour l'Automobile
Université Ibn Tofail, Kénitra | *2025 – 2027 (expected)*
Licence d'Excellence — Intelligence Artificielle et Ingénierie des Données
Université Ibn Tofail, Kénitra | *2025*
DUT — Génie Informatique
Université Ibn Tofail, Kénitra | *2024*
Baccalauréat — Sciences Physiques | 2022

PROFESSIONAL EXPERIENCE

Hyphen Group — AI Automation / Voice AI Intern | July 2026 – September 2026
- Contributed to AI automation projects with Voice AI agents, automated workflows, and AI-assisted business processes.
- Worked on an AI email-response workflow using historical email context, Retrieval-Augmented Generation (RAG), and scenario integration.

Acti-Sens — Scientific Research Intern | June 2026 – July 2026
- Worked on data analysis and Machine Learning modeling of EEG/fNIRS multimodal data related to cognitive variables.
- Contributed to model evaluation and the reproduction of scientific results to support research reproducibility.

Maison des Jeunes — Software Development / E-learning Intern | May 2024 – July 2024
- Contributed to software development and e-learning activities during an internship.

PROJECTS

DataPilot AI — Natural-Language Data Analytics Assistant | *Personal AI Project*
- Developed an application that lets nontechnical users upload structured datasets and ask questions in natural language, without SQL, Python, or BI tools; it converts questions into analytical operations and returns answers and charts.
- Target use cases: sales by country, top-10 products, monthly revenue trends, inactive customers, anomalous transactions, dataset insights.

CyberDetect AI — Intelligent Intrusion Detection System | *Academic Cybersecurity / Machine Learning Project*
- Developed a hybrid intrusion detection system combining XGBoost and LSTM (Deep Learning) to detect network attacks from traffic patterns in the CIC-IDS2017 dataset, with a real-time web interface and SHAP explainability (Explainable AI) for prediction interpretation and auditing. **Technologies:** Python, XGBoost, LSTM (PyTorch), SHAP, Pandas, NumPy, Flask, Scikit-learn

Telco Customer Churn Prediction and Retention System | *Academic Machine Learning Project*
- Built a Machine Learning system to predict customer churn and support retention decisions: data preprocessing, exploratory analysis, feature engineering, classifier comparison, cross-validation, class-imbalance handling, ROC-AUC, recall, and F1-score evaluation.

AgriSmart — Machine Learning for Moroccan Agriculture | *Academic Data Science / Machine Learning Project*
- Applied Machine Learning to crop analysis and agricultural decision-making in the Moroccan context: dataset preprocessing, exploratory analysis, crop classification, model evaluation, and visualization. **Technologies:** Python, Pandas, Scikit-learn, Random Forest

Smart Traffic Controller — Real-Time Traffic Light Control | *Academic Robotics / ROS2 Project*
- Designed a four-node ROS2 architecture (Sensor, Controller, Actuator, Dashboard) for a traffic-light control system simulated in Gazebo Classic with a TurtleBot3 controller; specified a strict 200 ms deadline and applied formal verification so conflicting lights cannot be green simultaneously. **Technologies:** ROS2 Humble, Python 3.10, rclpy, Gazebo Classic, TurtleBot3, Pandas, Matplotlib, NumPy

TECHNICAL SKILLS

Programming: Python, C/C++, Java, NetLogo | **Cybersecurity:** Intrusion Detection, Network Traffic Analysis
Machine Learning and Deep Learning: Regression, Classification, Clustering, Feature Engineering, Cross-Validation, Model Evaluation, Scikit-learn, XGBoost, MLP, RNN, LSTM, GRU, Transfer Learning, PyTorch, Explainable AI (SHAP)
Data Science and Visualization: Pandas, NumPy, Apache Spark, Hadoop, Power BI, Tableau, Plotly, Seaborn, Matplotlib, Streamlit
Generative AI and NLP: Transformers, Embeddings, Prompt Engineering, Retrieval-Augmented Generation (RAG)
AI Automation: n8n, Voice AI Agents, Automated Workflows | **Databases:** MySQL, NoSQL, PL/SQL, Data Modeling
Software Development and Systems: Laravel, Vue.js, PHP, JEE, REST APIs, Flask, Git, GitHub, Docker, ROS2 (Humble), rclpy, Gazebo Classic

CERTIFICATIONS AND TRAINING

Online training: Python, Pandas and Data Cleaning — Kaggle; Artificial Intelligence Fundamentals — IBM SkillsBuild; Create Machine Learning Models — Microsoft Learn; Data Visualization — Kaggle; AI Concepts for Developers — Microsoft Learn
Program participation: INJAZ Al-Arab program, in collaboration with PMI and PMIEF

LANGUAGES

French: Fluent | English: Fluent | Arabic: Native